

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-06

BOTANY

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Topics To Covered

- 1** Kingdom Protista (Slime Moulds, Protozoans)
- 2** Kingdom Fungi (Introduction)

SLIME MOULDS



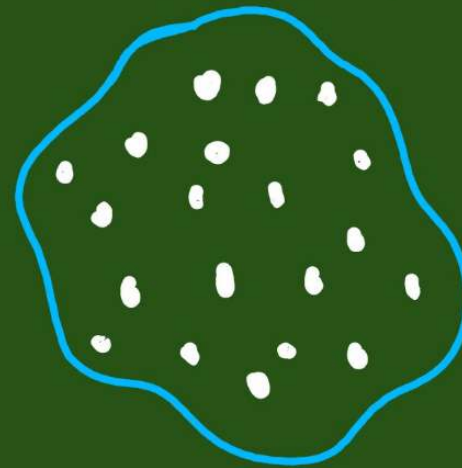
Habitat : Terristrial (soil)

• grow on decaying twigs and leaves (engulfs organic material).

Nutrition : Saprophytic (Slime moulds → Saprophytic protists).

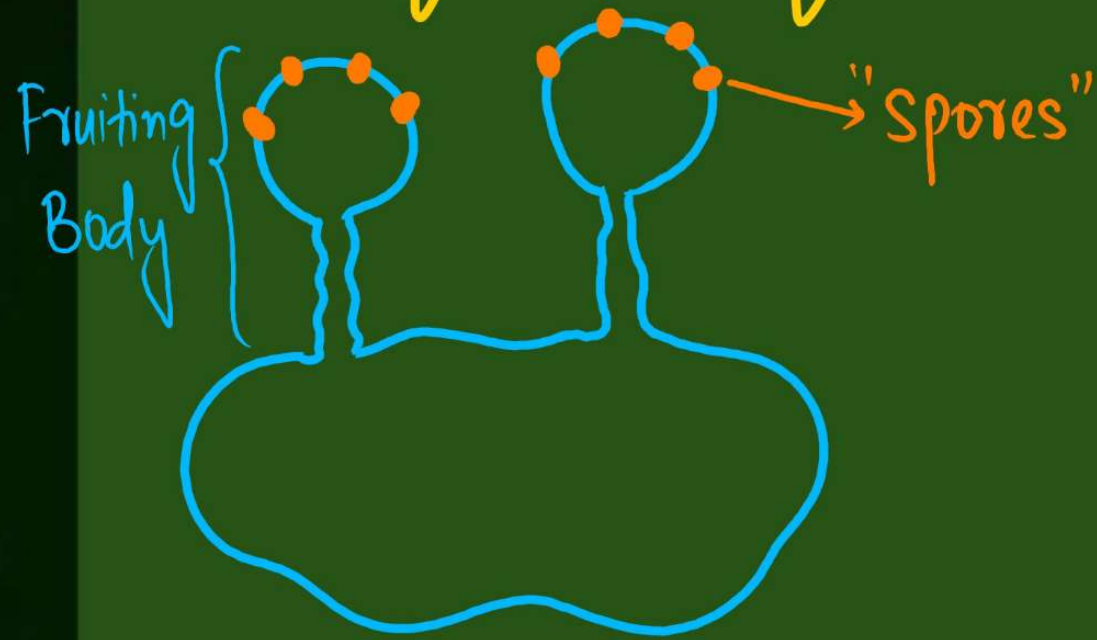
Cell wall : Absent. ✓

(*) Under favourable conditions, Slime moulds form an aggregation called as plasmodium.



Plasmodium

(*) under unfavourable conditions, plasmodium differentiates and forms fruiting body.



(*) fruiting body bears — spores at their tip.

Spores:

- ↳ possess true wall.
- ↳ Resistant to adverse conditions.
- ↳ Survive for years.
- ↳ dispersed by air currents.

PROTOZOANS

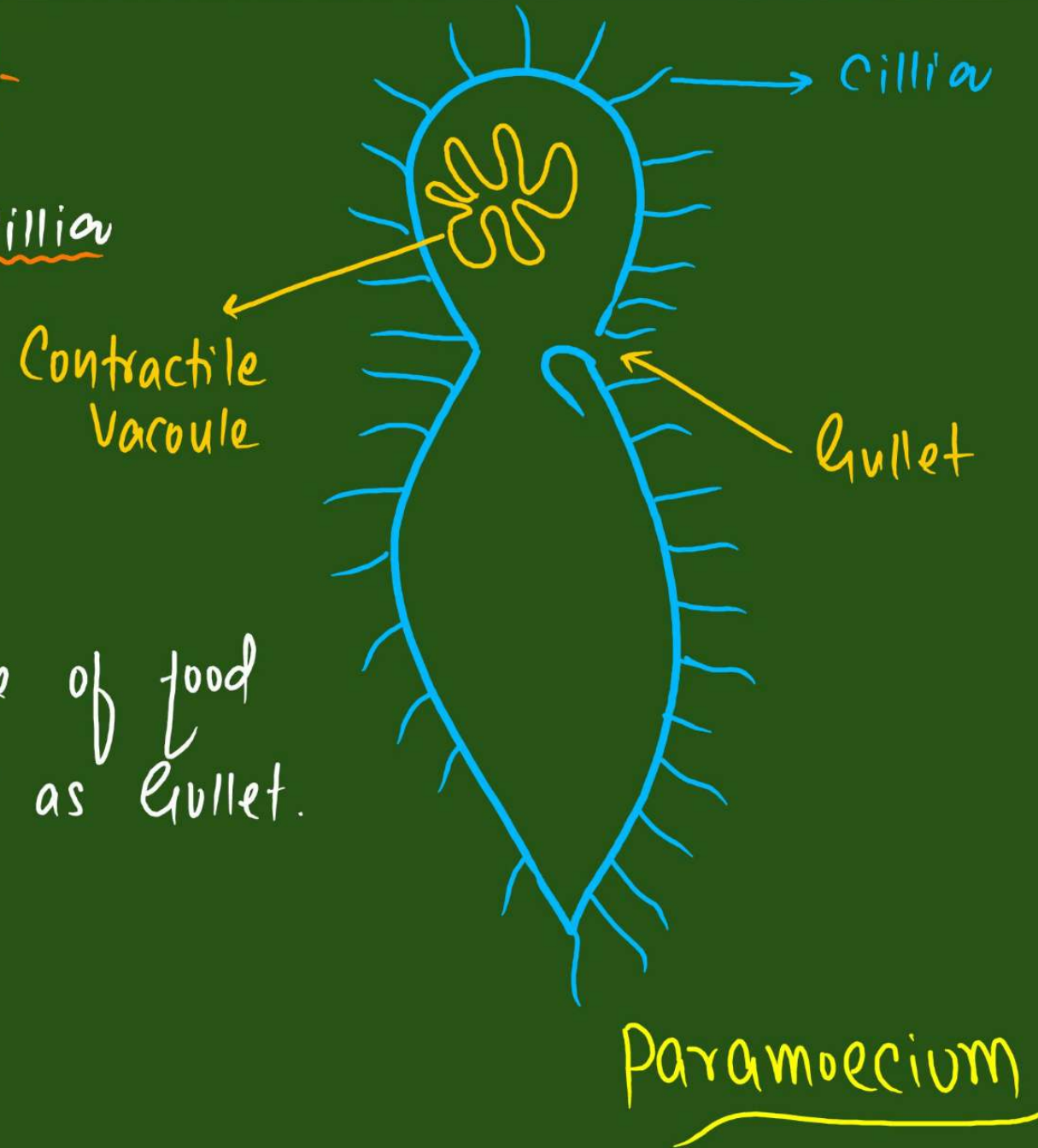


- Considered as primitive relatives of animals.
- Nutrition :- Heterotrophs (parasitic).
- Cell wall :- Absent.
- Protozoans are classified into 04 groups :-
 - ① Ciliated protozoans. ✓
 - ② Amoeboid protozoans. ✓
 - ③ Flagellated protozoans.
 - ④ Sporozoans

①. Ciliated Protozoans :-

- locomotory organ — is cilia
- Cilia — thousands in no.
- Cilia helps in locomotion.
AND
 also helps in intake of food through cavity kld as Gullet.

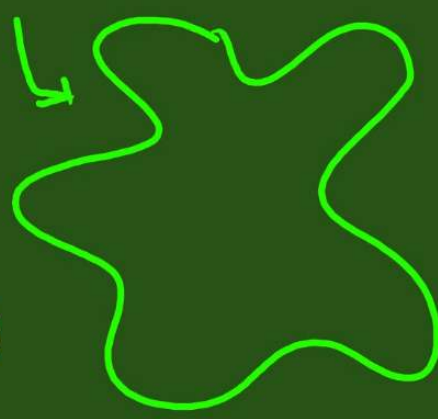
- Habitat :- → Aquatic
 → Few are parasitic.



② Amoeboid protozoans:-

- Habitat:-
 - Fresh H_2O
 - Marine H_2O (Sea H_2O)
 - Moist Soil.

pseudopodia



- ≡ Some are free living (Aquatic) ex. Amoeba
- ≡ " " parasitic ex. Entamoeba

↓
causes dysentery.

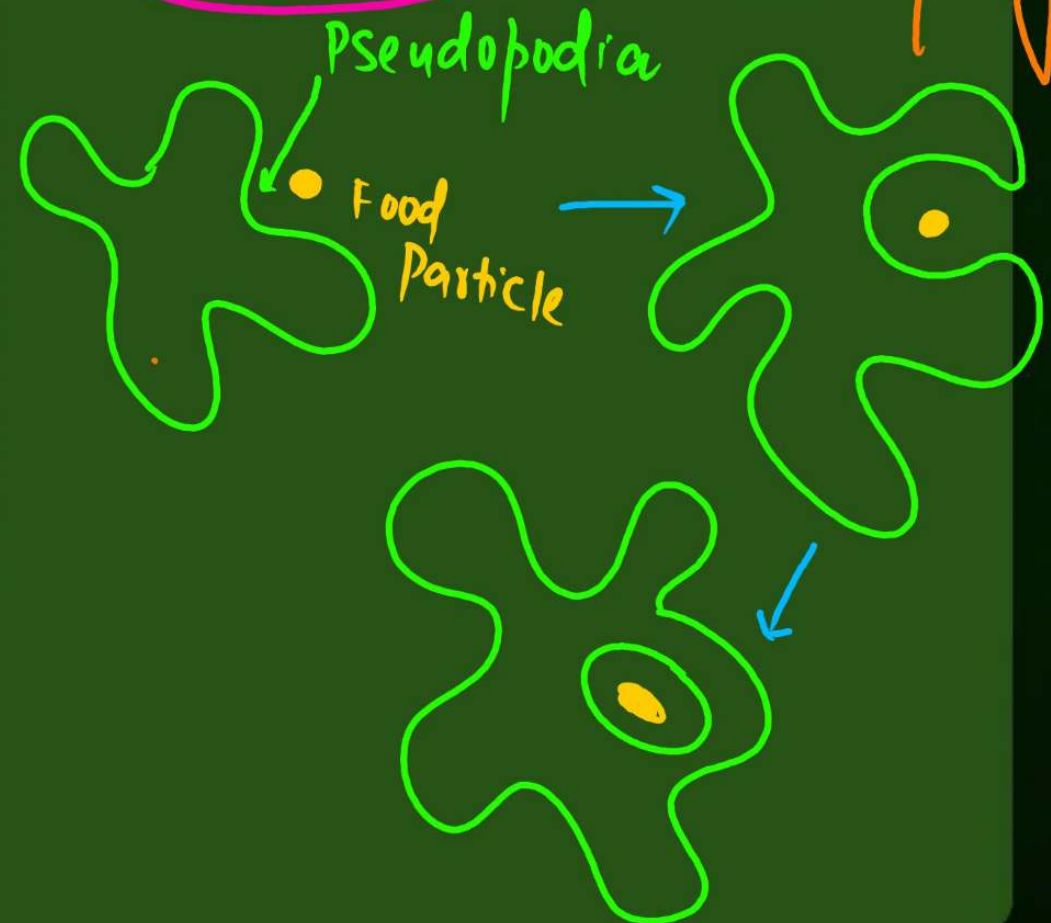
- ≡ Some marine forms — possess Silica Shells on their outer Surface

≡ They possess pseudopodia (false feet)

helps in locomotion

helps in capturing their prey.

pseudopodia



③ Flagellated protozoans :-

- Some are — free living (Aquatic)
- Some are — parasitic.
- locomotory organ — **Flagella**

Ex. Trypanosoma ↓
Causes Sleeping Sickness!

④ Sporozoans :-

- No locomotory organ.
- All Sporozoans are **endoparasites**.
↓
living inside host
- Sporozoans — possess infectious spore like stage in their life cycle

Ex. Plasmodium — malarial parasite
↓
Notorious (Famous for negative reasons)
Causes malaria in humans

• **Malaria** → has **staggering effect** (تیراں کن)
on human population

KINGDOM FUNGI

(Kingdom Mycota)



≡ It includes organisms which are :

a. achlorophyllous (without chlorophyll)

b. Eukaryotic.

c. Spore producing.

d. Thalloid organisms. (No differentiation of root, stem and leaf).

* Study of Fungi

is kld Mycology.

* Cell wall :- present

• Composed of "Chitin"

QUESTIONS

Which one of the following is wrong for fungi?

[2016 II]

- 1 They are heterotrophic ✓
- 2 They are both unicellular and multicellular ✓
Ex. Yeast
- 3 They are eukaryotic ✓
- ✓ 4 All fungi possess a purely cellulosic cell wall Chitin + ✗

QUESTIONS

4/10

Ciliates differ from all other protozoans in

[2018]

- 1 Using flagella for locomotion
- 2 Having a contractile vacuole for removing excess water
- 3 Using pseudopodia for capturing prey
- 4 Having two types of nuclei

QUESTIONS

H/w



Trypanosoma belongs to class:

[1989]

- 1 Sarcodina
- 2 Zooflagellata
- 3 Ciliata
- 4 Sporozoa

Thank
You